

Jinseok Bae

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Education

Ph.D. Seoul National University , Electrical and Computer Engineering • GPA: 3.98/4.30, <i>Advisor: Young Min Kim</i>	Sep 2022 – Feb 2027 (expected)
M.S. Seoul National University , Electrical and Computer Engineering • GPA: 4.18/4.30, <i>Advisor: Young Min Kim</i>	Mar 2020 – Feb 2022
B.S. Seoul National University , Biosystems Engineering, Electrical and Computer Engineering • GPA: 3.90/4.30	Mar 2014 – Feb 2020

Experience

Meta Reality Labs , Research Scientist Intern • Momentum Team, Physics AI (Gemini), <i>Manager: Yuting Ye</i>	Redmond, WA, USA Aug 2025 – Feb 2026
Roblox , Research Intern • Core AI, Game Engine Department, <i>Manager: Mubbasir Kapadia</i>	San Mateo, CA, USA Jun 2024 – Sep 2024
LG AI Research , Research Scientist • Vision AI	Seoul, Republic of Korea Jan 2022 – Jul 2022
Samsung Electronics , Undergraduate Intern • Health H/W Development Team, Mobile Division	Suwon, Republic of Korea Jan 2019 – Feb 2019
Teaching Assistant • Computer Graphics Programming (24'-fall) • 3D Computer Vision Track for AI Experts (Samsung, 21'-summer) • Signals and Systems (20'-spring) • Korean Course (for Keio University Exchange Students, 18'-summer)	
Academic Service • <i>Reviewer.</i> AAAI, ICCV, CVPR, SIGGRAPH, SIGGRAPH Asia, TVCG, Pacific Graphics	

Honors and Awards

Youlchon AI for All Fellowship Youlchon Foundation, Artificial Intelligence Institute at Seoul National University	26' spring
Honorable Mention, Changhae Young Researcher Award Korea Computer Graphics Society	KCGS 2025
Simulated Humanoid Wrestling Challenge (3rd Place) Team Yeti (Donggeun Lim, Minseok Kim, Jinseok Bae , Young Min Kim, Jungdam Won)	ICRA 2023
Merit-based Scholarship Seoul National University	19'-spring/fall 18'-fall
Agricultural Engineering Systems Scholarship Agricultural Engineering Systems Scholarship Foundation	18'-spring 15'-spring/fall

Publications

Project of {Motion Authoring, Centroidal Dynamics, Physics Simulation} <i>Jinseok Bae</i> , Yongseok Kim, Jeongseok Lee, Alexander Winkler, Yuting Ye, Young Min Kim	<i>Under Review</i> (SIGGRAPH Asia 2026)
Less is More: Improving Motion Diffusion Models with Sparse Keyframes <i>Jinseok Bae</i> , Inwoo Hwang, Young Yoon Lee, Ziyu Guo, Joseph Liu, Yizhak Ben-Shabat, Young Min Kim, Mubbasir Kapadia	ICCV 2025
Event-Driven Storytelling with Multiple Lifelike Humans in a 3D scene Donggeun Lim, <i>Jinseok Bae</i> , Inwoo Hwang, Seungmin Lee, Hwanhee Lee, Young Min Kim	ICCV 2025
Motion Synthesis with Sparse and Flexible Keyjoint Control Inwoo Hwang, <i>Jinseok Bae</i> , Donggeun Lim, Young Min Kim	ICCV 2025
PLT: Part-Wise Latent Tokens as Adaptable Motion Priors for Physically Simulated Characters <i>Jinseok Bae</i> , Younghwan Lee, Donggeun Lim, Young Min Kim	SIGGRAPH 2025
Goal-Driven Human Motion Synthesis in Diverse Tasks Inwoo Hwang, <i>Jinseok Bae</i> , Donggeun Lim, Young Min Kim	CVPR 2025 Workshop
Audio-aided Character Control for Inertial Measurement Tracking Hojun Jang, <i>Jinseok Bae</i> , Young Min Kim	Eurographics 2025 Short Paper
Versatile Physics-based Character Control with Hybrid Latent Representation <i>Jinseok Bae</i> , Jungdam Won, Donggeun Lim, Inwoo Hwang, Young Min Kim	Eurographics 2025
Dynamic Mesh Recovery from Partial Point Cloud Sequence Hojun Jang, Minkwan Kim, <i>Jinseok Bae</i> , Young Min Kim	ICCV 2023
PMP: Learning to Physically Interact with Environments using Part-wise Motion Priors <i>Jinseok Bae</i> , Jungdam Won, Donggeun Lim, Cheol-Hui Min, Young Min Kim	SIGGRAPH 2023
Neural Marionette: Unsupervised Learning of Motion Skeleton and Latent Dynamics from Volumetric Video <i>Jinseok Bae</i> , Hojun Jang, Cheol-Hui Min, Hyungun Choi, Young Min Kim	AAAI 2022 (Oral)
Auto-rigging 3D Bipedal Characters in Arbitrary Poses Jeonghwan Kim, Hyeontae Son, <i>Jinseok Bae</i> , Young Min Kim	Eurographics 2021 Short Paper
GATSBI: Generative Agent-centric Spatio-temporal Object Interaction Cheol-Hui Min, <i>Jinseok Bae</i> , Junho Lee, Young Min Kim	CVPR 2021 (Oral)